

Student Performance Analysis Using Python

Statistical Analysis and Data Visualization Report

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Abstract

This project performs a comprehensive statistical analysis of student performance data using Python. The analysis includes descriptive statistics, quartiles and percentiles, correlation analysis, probability calculations, normal distribution analysis, and outlier detection. Different visualization techniques such as scatter plots, histograms, and box plots are also used to better understand the relationships among variables.

The project demonstrates how statistical methods can be implemented programmatically to gain meaningful insights from educational datasets.

Objectives

The objectives of this project are:

- Analyze students' academic performance.
 - Calculate descriptive statistical measures.
 - Determine percentiles and quartiles.
 - Measure correlation between study-related factors and exam scores.
 - Calculate basic probabilities.
 - Examine score distribution using Z-scores.
 - Detect outliers using the Interquartile Range (IQR) method.
 - Visualize the results using graphs.
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Dataset Description

The dataset used is:

student_performance.csv

It contains information related to students' academic records.

Important attributes include:

Column	Description
Student_ID	Unique student identifier
Study_Hours	Number of study hours
Attendance	Attendance percentage
Assignment_Score	Assignment marks
Exam_Score	Final examination score

Software and Libraries Used

Library	Purpose
Pandas	Reading and manipulating datasets
NumPy	Numerical computations
Matplotlib	Data visualization
SciPy (stats)	Statistical functions

Methodology

The project follows six major statistical tasks.

Task 1: Descriptive Statistics

Purpose

Descriptive statistics summarize the central tendency and spread of the exam scores.

Calculated Measures

- Mean

- Median
- Mode
- Variance
- Standard Deviation

Python Functions Used

- `mean()`
 - `median()`
 - `mode()`
 - `var()`
 - `std()`
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Task 2: Percentiles and Quartiles

Purpose

Quartiles divide the data into four equal parts.

Calculated Values

- Minimum
- Q1 (25%)
- Q2 (Median)
- Q3 (75%)
- Maximum

Python Function

```
quantile()
```

Interpretation

- Q1 represents the lower 25% of students.
- Q2 represents the median score.
- Q3 represents the top 25% boundary.

These values help understand score distribution and are later used for outlier detection.

Task 3: Correlation Analysis

Purpose

Correlation measures the strength of the relationship between variables.

The project computes the correlation between:

- Study Hours and Exam Score
- Attendance and Exam Score
- Assignment Score and Exam Score

Python Function

```
corr()
```

Visualization

A scatter plot is generated between:

- Study Hours
- Exam Score

Interpretation

Higher positive correlation indicates that increasing study hours generally leads to better examination performance.

Task 4: Probability Basics

Purpose

Simple probabilities are calculated from the dataset.

Examples include:

- Probability of passing the exam
- Probability of scoring above 80
- Probability of studying more than five hours

Interpretation

Probability values provide insight into student success rates and study behavior.

Task 5: Normal Distribution

Purpose

To determine whether exam scores approximately follow a normal distribution.

Z-Score Calculation

Each student's score is converted into a Z-score.

Visualization

Histogram of Exam Scores

The histogram illustrates the distribution of scores across students.

Task 6: Outlier Detection

Purpose

Detect unusually high or low exam scores.

Method

Interquartile Range (IQR)

Formula

$$IQR = Q3 - Q1$$

Lower Bound

$$Q1 - 1.5(IQR)$$

Upper Bound

$$Q3 + 1.5(IQR)$$

Any score outside these limits is classified as an outlier.

Visualization

A box plot is generated to display

- Median
 - Quartiles
 - Whiskers
 - Outliers
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Visualizations Produced

The notebook generates the following graphical outputs:

Scatter Plot

Shows the relationship between:

- Study Hours
- Exam Score

Purpose:

Identify whether increased study time improves exam performance.

Histogram

Displays the frequency distribution of exam scores.

Purpose:

Evaluate whether scores follow a normal distribution.

Box Plot

Displays

- Median
- Quartiles
- Outliers

Purpose:

Detect extreme values in the dataset.

Results

The analysis successfully computes:

- Measures of central tendency
- Measures of dispersion
- Quartiles
- Percentiles
- Correlation coefficients
- Student pass probabilities
- Z-scores
- IQR outliers

The visualizations further improve understanding of student academic performance.

Applications

This analysis can be applied in:

- Educational institutions
 - Student performance monitoring
 - Academic research
 - Learning analytics
 - Performance prediction systems
 - Educational data mining
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Conclusion

The project demonstrates the practical implementation of statistical analysis using Python. By combining descriptive statistics, probability theory, correlation analysis, normal distribution, and outlier detection, the project provides meaningful insights into student performance. The visualizations complement the numerical analysis and make interpretation easier for educators and researchers.
